[PRD] Low-Cost AI Spirometer & Primary Respiratory Screening Solution

SpiroAI Product Requirements Document

1. Executive Summary

- **Product Name:** SpiroAI (working title)
- **Product Definition:** A **smart respiratory screening solution** that combines low-cost hardware sensors with an AI error-correction algorithm, enabling accurate early screening for pulmonary disease (COPD, asthma) at low cost even in primary health centers lacking specialized medical equipment.
- **Core Problem:** Addressing the high late-stage diagnosis rate (**48.4%**) and resulting mortality of pulmonary disease and lung cancer caused by an initially asymptomatic course.

> **"Deploy $30-50 hardware at 1/20th the price of clinical-grade equipment, and use AI algorithms to achieve clinical-screening-level accuracy — closing the healthcare gap in remote areas and primary health centers."**

2. Situation Analysis & Problem Statement

2.1 Target Region & Personas (Pain Points)

- **Geographic/Economic Target:** Primary health center (FAP) medical staff and local residents in Kyrgyzstan and the broader Central Asian LMIC region.
- **Key Stakeholders Affected:**
  1. **Primary health center staff:** Unable to adopt high-cost diagnostic equipment ($1,000-$3,000+) → no option but to transfer patients to large hospitals in major cities.
  2. **Local residents (patients):** Often unaware of respiratory illness due to its asymptomatic early stage, and delay seeking care due to transportation and treatment costs → present only after the condition has severely worsened.
  3. **Public health authorities:** Budgets for affordable equipment are grossly insufficient relative to high disease prevalence.

2.2 Data Evidence

- **Air Pollution & Prevalence:** Major cities such as Bishkek record some of the world's highest fine particulate matter levels in winter due to coal heating and smog.
- **Healthcare Infrastructure:** Availability of specialized medical staff and diagnostic equipment per 100,000 population is among the lowest in the world; over 90% of rural primary health centers (FAPs) lack a spirometer.

- **Diagnostic Delay Indicator:** The late-stage diagnosis rate for pulmonary disease and lung cancer stands at 48.4%, a direct consequence of the lack of early-screening infrastructure.

2.3 Limitations of Existing Solutions (Market Gap)

- **High-cost existing equipment:** PC-based spirometers and clinical devices designed for developed markets cost hundreds to thousands of dollars (roughly 80,000-300,000+ som), making adoption in low-resource countries unfeasible.
- **Limitations of mobile apps:** Smartphone-microphone-based voice estimation is not a physical measurement, so reliability is poor.
- **Lack of trained personnel:** Rural and semi-urban environments have an acute shortage of trained technicians able to administer spirometry tests.

3. Core Functional Requirements

Module	Functional Requirement	Description
HW (Device)	Low-cost flow measurement	Collects real flow and volume measurement data using an ESP32 MCU (with BLE) and a low-cost differential-pressure/flow sensor.
HW (Hygiene)	Disposable BVF filter	Incorporates a disposable Bacterial/Viral Filter (BVF) that captures 99.999% of microorganisms, eliminating cross-infection risk at the source.
AI (Algorithm)	1D-CNN error-correction model	Precisely corrects the mechanical error and non-linear noise of low-cost sensors using advanced AI models such as 1D-CNN, achieving clinical-grade validity.
SW (Sensor Fusion)	Digital multi-sensor fusion	Integrates device measurement data with local real-time weather API data (temperature, humidity, particulate matter) to correct for environmental variables.
SW (App)	Real-time measurement guidance	Provides in-app animation and native-language voice coaching in real time to guide users toward a correct, valid test without requiring a professional.
SW (Results)	3-tier risk classification	Calculates FEV1 and FVC values and presents an intuitive 3-tier result: [Normal / Monitor / Refer to Specialist].
SW (Integration)	Cloud sync & EHR integration	Automatically syncs measurement data with public health network Electronic Health Records (EHR) to serve as a data foundation for remote care.

4. Competitive Advantage

Comparison	Hospital-Grade Clinical Device	Mobile App (Voice-Only)	SpiroAI (This Product)
Price	$1,000 - $3,000+ (very high)	Free - $5	**$30 - $50 (low-cost)**
Measurement method	Precision sensor-based physical measurement	Smartphone microphone voice estimation	**Low-cost sensor physical measurement + 1D-CNN AI correction**
Accuracy	Highest (diagnostic-grade)	Low (poor reliability)	**Moderate-to-high (screening-grade)**
Setting	Limited to advanced hospitals	Personal use	**Rural areas/primary health centers, mobile outreach clinics**

5. Core Technology & Differentiation

- **AI error-correction algorithm:** Advanced AI models such as 1D-CNN precisely correct the mechanical error of low-cost sensors, achieving a valid correlation of 85-90%+ relative to clinical equipment.
- **Digital multi-sensor fusion:** Integrating device data with local real-time weather API data (temperature, humidity, particulate matter) enhances measurement accuracy and risk interpretation.
- **Overwhelming price competitiveness:** 95% cheaper than hospital-grade devices, with a $30-$50 target price optimized for B2B/B2G procurement.
- **Cloud synchronization:** Automatic sync with public health network EHRs provides a data foundation for remote care.

6. Technical & Economic Feasibility

6.1 Technical Feasibility

- **Component composition (BOM-based cost reduction):** Main MCU ESP32 (~$3), low-cost differential-pressure/flow sensor (~$5), battery and case/mouthpiece (~$7) → **target production cost of $15-$20**
- **Accuracy strategy:** The non-linear output characteristics of low-cost sensors are precisely corrected via a 1D-CNN-based AI regression model, achieving a valid correlation of 85-90%+ relative to clinical equipment.
- **Target positioning:** Rather than replacing high-cost diagnostic equipment, the product targets "primary screening," enabling a fast and complete algorithmic implementation.

6.2 Economic Feasibility

- **Target selling price:** $30-$50 (95%+ cost reduction versus existing clinical equipment)
- **Margin at scale:** Maintaining a cost ratio below 40% at scale secures a stable margin for B2B/B2G supply.

7. Go-to-Market Strategy (GTM & TAM-SAM-SOM)

```
[TAM] $1.5B  Central Asia & LMIC primary medical device & respiratory screening market (currently 70% c
   └── [SAM] $120M  Primary health centers (FAPs) & corporate health clinics in Kyrgyzstan and 5 neighb
          └── [SOM] $500K  1 partner university health center + 5 pilot health centers in Kyrgyzstan (B
```

1. **SOM (Beachhead entry):** Pilot rollout to 1 partner university health center and 5 pilot health centers to gather field validation data.
2. **SAM (Expansion):** B2B rollout across Kyrgyzstan's roughly 1,000 primary health centers (FAPs) and mining/manufacturing firms with high air-pollution exposure.
3. **TAM (Global):** Following field validation at the beachhead, expand into 5 Central Asian countries and other LMICs in partnership with KOICA ODA and WHO health programs.

8. SWOT Analysis

Strengths

- Ultra-low cost: A low-cost hardware design with a BOM of roughly $20 completely removes the price barrier.
- Hygiene and safety: A disposable BVF filter eliminates cross-infection risk when the device is shared.
- User-friendliness: Real-time native-language in-app coaching enables self-administered testing without a medical professional.

Weaknesses

- Hardware precision limits: Low-cost flow sensors have inherent physical error (though this is overcome to clinical-grade levels via the 1D-CNN AI correction algorithm).
- Infrastructure dependency: App connectivity and cloud transmission require a local smartphone and network environment.

Opportunities

- Surging demand: Severe winter smog and air pollution are driving a sharp rise in demand for respiratory screening across Central Asia and other LMIC countries.
- Public partnerships: Easy market entry (B2G) via pilot programs with local university health centers and primary health centers (FAPs) in Kyrgyzstan.
- Global scalability: Partnership with KOICA ODA and WHO health programs can secure

Threats

- Initial entry barriers: Early psychological/infrastructural resistance to adopting digital medical devices and cloud systems in rural areas.
- Competitive dynamics: Possibility of established global spirometer manufacturers (US/Europe) launching low-cost models targeting emerging markets.

expansion infrastructure into 5 neighboring countries.	

9. Business Model & Sustainability

- **Razor-Blade model (core recurring revenue):**
  - Razor: Low-cost device unit ($30-$50) minimizes the entry barrier.
  - Blade: Ongoing subscription and delivery of the disposable mouthpiece (BVF antimicrobial filter) consumed at every patient test, generating **recurring, sustainable revenue**.
- **SaaS subscription billing:** Monthly/annual access fees for an "institutional dashboard" for centralized cloud management of measurement data.
- **B2B/B2G revenue diversification:** Sale of licenses for periodic screening of mining/manufacturing workers and public health project (ODA) contract wins.

10. Conclusion & Expected Impact

- **Closing the healthcare gap:** Low-cost hardware plus AI correction technology overcomes the structural limitations of primary health centers in resource-poor settings.
- **Improved national health indicators:** Continuous remote data tracking enables early detection of respiratory disease and pre-emptive prevention of late-stage cancer progression.
- **Sustainability:** SaaS subscriptions and recurring consumable revenue together achieve both social impact and business profitability.